

```
In [ ]: """  
Enter your full name here: Elijah Oliver  
"""
```

```
In [1]: #Run this cell first before moving on  
import pandas as pd  
import matplotlib.pyplot as plt  
import seaborn as sns  
plt.style.use("ggplot")  
%matplotlib inline
```

```
In [4]: #Use the variables/urls to load the data  
# url for covid cases:  
covid_cases = "https://github.com/CSSEGISandData/COVID-19/raw/master/c  
  
# url for covid deaths:  
covid_deaths = "https://github.com/CSSEGISandData/COVID-19/raw/master/  
  
# It is okay to have a horizontal scroll bar for this cell.  
df_cases = pd.read_csv(covid_cases)  
df_deaths = pd.read_csv(covid_deaths) # include covid_deaths variable
```

In [6]: *#Run the cell, review the output, and move to the next cell*
df_cases

Out [6]:

	UID	iso2	iso3	code3	FIPS	Admin2	Province_State	Country_Region	
0	84001001	US	USA	840	1001.0	Autauga	Alabama	US	32.5%
1	84001003	US	USA	840	1003.0	Baldwin	Alabama	US	30.7%
2	84001005	US	USA	840	1005.0	Barbour	Alabama	US	31.8%
3	84001007	US	USA	840	1007.0	Bibb	Alabama	US	32.9%
4	84001009	US	USA	840	1009.0	Blount	Alabama	US	33.9%
...	...	...	...	...	...	...	...	...	...
3337	84056039	US	USA	840	56039.0	Teton	Wyoming	US	43.9%
3338	84056041	US	USA	840	56041.0	Uinta	Wyoming	US	41.2%
3339	84090056	US	USA	840	90056.0	Unassigned	Wyoming	US	0.0%
3340	84056043	US	USA	840	56043.0	Washakie	Wyoming	US	43.9%
3341	84056045	US	USA	840	56045.0	Weston	Wyoming	US	43.8%

3342 rows × 1154 columns

In [7]: *#Run the cell, review the output, and move to the next cell*
df_deaths

Out [7]:

	UID	iso2	iso3	code3	FIPS	Admin2	Province_State	Country_Region	
0	84001001	US	USA	840	1001.0	Autauga	Alabama	US	32.53
1	84001003	US	USA	840	1003.0	Baldwin	Alabama	US	30.72
2	84001005	US	USA	840	1005.0	Barbour	Alabama	US	31.86
3	84001007	US	USA	840	1007.0	Bibb	Alabama	US	32.99
4	84001009	US	USA	840	1009.0	Blount	Alabama	US	33.98
...	...	...	...	...	...	...	...	...	...
3337	84056039	US	USA	840	56039.0	Teton	Wyoming	US	43.93
3338	84056041	US	USA	840	56041.0	Uinta	Wyoming	US	41.28
3339	84090056	US	USA	840	90056.0	Unassigned	Wyoming	US	0.00
3340	84056043	US	USA	840	56043.0	Washakie	Wyoming	US	43.90
3341	84056045	US	USA	840	56045.0	Weston	Wyoming	US	43.83

3342 rows × 1155 columns

In [8]: *"""Problem 0"""*
Run the cell, notice the additional data column in df_deaths which w
print("Cases DataFrame:", list(df_cases.columns[0:12]))
print("") # Do not modify - this displays blank line for space between
print("Deaths DataFrame:", list(df_deaths.columns[0:13]))

Cases DataFrame: ['UID', 'iso2', 'iso3', 'code3', 'FIPS', 'Admin2', 'Province_State', 'Country_Region', 'Lat', 'Long_', 'Combined_Key', '1/22/20']

Deaths DataFrame: ['UID', 'iso2', 'iso3', 'code3', 'FIPS', 'Admin2', 'Province_State', 'Country_Region', 'Lat', 'Long_', 'Combined_Key', 'Population', '1/22/20']

```
In [15]: """Problem 1a"""
#Remove the following columns for df_cases using the drop method:
# "UID", "iso2", "iso3", "code3", "FIPS", "Country_Region",
# "Lat", "Long_"
# Use the inplace=True.

df_cases.drop(columns=["UID", "iso2", "iso3", "code3", "FIPS", "Country_Region"], inplace=True)
df_cases.head()
```

Out[15]:

	1/27/20	1/28/20	...	2/28/23	3/1/23	3/2/23	3/3/23	3/4/23	3/5/23	3/6/23	3/7/23	3/8/23	3/9/23
0	0	0	...	19732	19759	19759	19759	19759	19759	19759	19759	19790	19790
0	0	0	...	69641	69767	69767	69767	69767	69767	69767	69767	69860	69860
0	0	0	...	7451	7474	7474	7474	7474	7474	7474	7474	7485	7485
0	0	0	...	8067	8087	8087	8087	8087	8087	8087	8087	8091	8091
0	0	0	...	18616	18673	18673	18673	18673	18673	18673	18673	18704	18704

```
In [16]: """Problem 1b"""
#Remove the same columns for df_deaths but also include the extra data
df_deaths.drop(columns=["UID", "iso2", "iso3", "code3", "FIPS", "Country_Region"], inplace=True)
df_deaths.tail(3)
```

Out[16]:

	Admin2	Province_State	Combined_Key	1/22/20	1/23/20	1/24/20	1/25/20	1/26/20
3339	Unassigned	Wyoming	Unassigned, Wyoming, US	0	0	0	0	0
3340	Washakie	Wyoming	Washakie, Wyoming, US	0	0	0	0	0
3341	Weston	Wyoming	Weston, Wyoming, US	0	0	0	0	0

3 rows × 1146 columns

```
In [24]: """Problem 2"""
#Rename the following columns for both dataframes using the rename method
# Admin2 to county, Province_State to state, Combined_key to county_state
print("Before rename:", list(df_cases.columns)) # displays data column names
df_cases.rename(columns={"Admin2": 'county', "Province_State": 'state', "Combined_Key": 'county_state'})
```

```

print("After rename:", list(df_cases.columns))

print("")

print("Before rename:", list(df_deaths.columns)) # displays data column
df_deaths.rename(columns={"Admin2": 'county', "Province_State": 'state', "
print("After rename:", list(df_deaths.columns)) # displays data column

print("")

```

```

Before rename: ['county', 'state', 'Combined_Key', '1/22/20', '1/23/20', '1/24/20', '1/25/20', '1/26/20', '1/27/20', '1/28/20', '1/29/20', '1/30/20', '1/31/20', '2/1/20', '2/2/20', '2/3/20', '2/4/20', '2/5/20', '2/6/20', '2/7/20', '2/8/20', '2/9/20', '2/10/20', '2/11/20', '2/12/20', '2/13/20', '2/14/20', '2/15/20', '2/16/20', '2/17/20', '2/18/20', '2/19/20', '2/20/20', '2/21/20', '2/22/20', '2/23/20', '2/24/20', '2/25/20', '2/26/20', '2/27/20', '2/28/20', '2/29/20', '3/1/20', '3/2/20', '3/3/20', '3/4/20', '3/5/20', '3/6/20', '3/7/20', '3/8/20', '3/9/20', '3/10/20', '3/11/20', '3/12/20', '3/13/20', '3/14/20', '3/15/20', '3/16/20', '3/17/20', '3/18/20', '3/19/20', '3/20/20', '3/21/20', '3/22/20', '3/23/20', '3/24/20', '3/25/20', '3/26/20', '3/27/20', '3/28/20', '3/29/20', '3/30/20', '3/31/20', '4/1/20', '4/2/20', '4/3/20', '4/4/20', '4/5/20', '4/6/20', '4/7/20', '4/8/20', '4/9/20', '4/10/20', '4/11/20', '4/12/20', '4/13/20', '4/14/20', '4/15/20', '4/16/20', '4/17/20', '4/18/20', '4/19/20', '4/20/20', '4/21/20', '4/22/20', '4/23/20', '4/24/20', '4/25/20', '4/26/20', '4/27/20', '4/28/20', '4/29/20', '4/30/20', '5/1/20', '5/2/20', '5/3/20', '5/4/20', '5/5/20', '5/6/20', '5/7/20', '5/8/20', '5/9/20', '5/10/20', '5/11/20', '5/12/20', '5/13/20', '5/14/20', '5/15/20', '5/16/20', '5/17/20', '5/18/20', '5/19/20', '5/20/20', '5/21/20', '5/22/20', '5/23/20', '5/24/20', '5/25/20', '5/26/20', '5/27/20', '5/28/20', '5/29/20', '5/30/20', '5/31/20', '6/1/20', '6/2/20', '6/3/20', '6/4/20', '6/5/20', '6/6/20', '6/7/20', '6/8/20', '6/9/20', '6/10/20', '6/11/20', '6/12/20', '6/13/20', '6/14/20', '6/15/20', '6/16/20', '6/17/20', '6/18/20', '6/19/20', '6/20/20', '6/21/20', '6/22/20', '6/23/20', '6/24/20', '6/25/20', '6/26/20', '6/27/20', '6/28/20', '6/29/20', '6/30/20', '7/1/20', '7/2/20', '7/3/20', '7/4/20', '7/5/20', '7/6/20', '7/7/20', '7/8/20', '7/9/20', '7/10/20', '7/11/20', '7/12/20', '7/13/20', '7/14/20', '7/15/20', '7/16/20', '7/17/20', '7/18/20', '7/19/20', '7/20/20', '7/21/20', '7/22/20', '7/23/20', '7/24/20', '7/25/20', '7/26/20', '7/27/20', '7/28/20', '7/29/20', '7/30/20', '7/31/20', '8/1/20', '8/2/20', '8/3/20', '8/4/20', '8/5/20', '8/6/20', '8/7/20', '8/8/20', '8/9/20', '8/10/20', '8/11/20', '8/12/20', '8/13/20', '8/14/20', '8/15/20', '8/16/20', '8/17/20', '8/18/20', '8/19/20', '8/20/20', '8/21/20', '8/22/20', '8/23/20', '8/24/20', '8/25/20', '8/26/20', '8/27/20', '8/28/20', '8/29/20', '8/30/20', '8/31/20', '9/1/20', '9/2/20', '9/3/20', '9/4/20', '9/5/20', '9/6/20', '9/7/20', '9/8/20', '9/9/20', '9/10/20', '9/11/20', '9/12/20', '9/13/20', '9/14/20', '9/15/20', '9/16/20', '9/17/20', '9/18/20', '9/19/20', '9/20/20', '9/21/20', '9/22/20', '9/23/20', '9/24/20', '9/25/20', '9/26/20', '9/27/20', '9/28/20', '9/29/20', '9/30/20', '10/1/20', '10/2/20', '10/3/20', '10/4/20', '10/5/20', '10/6/20', '10/7/20', '10/8/20', '10/9/20', '10/10/20', '10/11/20', '10/12/20', '10/13/20', '10/14/20', '10/15/20', '10/16/20', '10/17/20', '10/18/20', '10/19/20', '10/20/20', '10/21/20', '10/22/20', '10/23/20', '10/24/20', '10/25/20', '10/26/20', '10/27/20', '10/28/20', '10/29/20', '10/30/20', '10/31/20', '11/1/20', '11/2/20', '11/3/20', '11/4/20', '11/5/20', '11/6/20', '11/7/20', '11/8/20', '11/9/20', '11/10/20', '11/11/20', '11/12/20', '11/13/20', '11/14/20', '11/15/20', '11/16/20', '11/17/20', '11/18/20', '11/19/20', '11/20/20', '11/21/20', '11/22/20', '11/23/20', '11/24/20', '11/25/20', '11/26/20', '11/27/20', '11/28/20', '11/29/20', '11/30/20', '12/1/20', '12/2/20', '12/3/20', '12/4/20', '12/5/20', '12/6/20', '12/7/20', '12/8/20', '12/9/20', '12/10/20', '12/11/20', '12/12/20', '12/13/20', '12/14/20', '12/15/20', '12/16/20', '12/17/20', '12/18/20', '12/19/20', '12/20/20', '12/21/20', '12/22/20', '12/23/20', '12/24/20', '12/25/20', '12/26/20', '12/27/20', '12/28/20', '12/29/20', '12/30/20', '12/31/20']

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[illegible]

	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045	2046	2047	2048	2049	2050	2051	2052	2053	2054	2055	2056	2057	2058	2059	2060	2061	2062	2063	2064	2065	2066	2067	2068	2069	2070	2071	2072	2073	2074	2075	2076	2077	2078	2079	2080	2081	2082	2083	2084	2085	2086	2087	2088	2089	2090	2091	2092	2093	2094	2095	2096	2097	2098	2099	2100	2101	2102	2103	2104	2105	2106	2107	2108	2109	2110	2111	2112	2113	2114	2115	2116	2117	2118	2119	2120	2121	2122	2123	2124	2125	2126	2127	2128	2129	2130	2131	2132	2133	2134	2135	2136	2137	2138	2139	2140	2141	2142	2143	2144	2145	2146	2147	2148	2149	2150	2151	2152	2153	2154	2155	2156	2157	2158	2159	2160	2161	2162	2163	2164	2165	2166	2167	2168	2169	2170	2171	2172	2173	2174	2175	2176	2177	2178	2179	2180	2181	2182	2183	2184	2185	2186	2187	2188	2189	2190	2191	2192	2193	2194	2195	2196	2197	2198	2199	2200	2201	2202	2203	2204	2205	2206	2207	2208	2209	2210	2211	2212	2213	2214	2215	2216	2217	2218	2219	2220	2221	2222	2223	2224	2225	2226	2227	2228	2229	2230	2231	2232	2233	2234	2235	2236	2237	2238	2239	2240	2241	2242	2243	2244	2245	2246	2247	2248	2249	2250	2251	2252	2253	2254	2255	2256	2257	2258	2259	2260	2261	2262	2263	2264	2265	2266	2267	2268	2269	2270	2271	2272	2273	2274	2275	2276	2277	2278	2279	2280	2281	2282	2283	2284	2285	2286	2287	2288	2289	2290	2291	2292	2293	2294	2295	2296	2297	2298	2299	2300	2301	2302	2303	2304	2305	2306	2307	2308	2309	2310	2311	2312	2313	2314	2315	2316	2317	2318	2319	2320	2321	2322	2323	2324	2325	2326	2327	2328	2329	2330	2331	2332	2333	2334	2335	2336	2337	2338	2339	2340	2341	2342	2343	2344	2345	2346	2347	2348	2349	2350	2351	2352	2353	2354	2355	2356	2357	2358	2359	2360	2361	2362	2363	2364	2365	2366	2367	2368	2369	2370	2371	2372	2373	2374	2375	2376	2377	2378	2379	2380	2381	2382	2383	2384	2385	2386	2387	2388	2389	2390	2391	2392	2393	2394	2395	2396	2397	2398	2399	2400	2401	2402	2403	2404	2405	2406	2407	2408	2409	2410	2411	2412	2413	2414	2415	2416	2417	2418	2419	2420	2421	2422	2423	2424	2425	2426	2427	2428	2429	2430	2431	2432	2
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```

```
In [25]: """Problem 3"""
Melt/reshape df_cases. Columns to assign to
# the id_vars parameter are: county, state, and county_state.
Make sure that var_name="dates" and value_name="cases"
Complete the melt function code below.
f_cases_melted = pd.melt(df_cases, id_vars=["county", "state", "county_
f_cases_melted.tail(3)
```

Out[25]:

	county	state	county_state	dates	cases
3819903	Unassigned	Wyoming	Unassigned, Wyoming, US	3/9/23	0
3819904	Washakie	Wyoming	Washakie, Wyoming, US	3/9/23	2755
3819905	Weston	Wyoming	Weston, Wyoming, US	3/9/23	1905

```
In [27]: """Problem 4"""
Melt/reshape df_deaths. Columns to assign to
# the id_vars parameter are: county, state, and county_state.
Make sure that var_name="dates" and value_name="deaths"
Complete the melt function code below.

deaths_melted = pd.melt(df_deaths, id_vars=["county", "state", "county_s
deaths_melted.tail(3)
```

Out[27]:

	county	state	county_state	dates	deaths
3819903	Unassigned	Wyoming	Unassigned, Wyoming, US	3/9/23	0
3819904	Washakie	Wyoming	Washakie, Wyoming, US	3/9/23	51
3819905	Weston	Wyoming	Weston, Wyoming, US	3/9/23	23

```
In [30]: """Problem 5"""
#Merge the melted dataframes for cases and deaths.
#Complete the merged function code below.

df_merged = pd.merge(df_cases_melted, df_deaths_melted,
                      how = "left", on = ["county", "state", "county_state"])
df_merged.tail()
```

Out[30]:

	county	state	county_state	dates	cases	deaths
3819901	Teton	Wyoming	Teton, Wyoming, US	3/9/23	12134	16
3819902	Uinta	Wyoming	Uinta, Wyoming, US	3/9/23	6406	43
3819903	Unassigned	Wyoming	Unassigned, Wyoming, US	3/9/23	0	0
3819904	Washakie	Wyoming	Washakie, Wyoming, US	3/9/23	2755	51
3819905	Weston	Wyoming	Weston, Wyoming, US	3/9/23	1905	23

```
In [34]: """Problem 6"""
#Using df_merged, change data type for dates to a datetime object.

df_merged.dates = pd.to_datetime(df_merged["dates"])
df_merged.dtypes
```

```
/var/folders/1_/x7h8qgkn6pxg12s42yd43yfh0000gn/T/ipykernel_19921/2170
152627.py:4: UserWarning: Could not infer format, so each element will
be parsed individually, falling back to `dateutil`. To ensure parsing
is consistent and as-expected, please specify a format.
df_merged.dates = pd.to_datetime(df_merged["dates"])
```

```
Out[34]: county          object
state          object
county_state    object
dates          datetime64[ns]
cases          int64
deaths         int64
dtype: object
```

```
In [36]: """Problem 7"""
#Calculate the number of days into the outbreak for each record.
#The name of the new column will be us_outbreak

df_merged["us_outbreak"] = (df_merged['dates'] - df_merged['dates'].min()) / (df_merged[["dates", "us_outbreak"]].iloc[-1] - df_merged[["dates", "us_outbreak"]].iloc[0]) # this code outputs the n
```

```
Out[36]: dates          2023-03-09 00:00:00
us_outbreak          1142
Name: 3819905, dtype: object
```

```
In [37]: #Run cell, review the dataframe output which include the new column (us_outbreak)
df_merged
```

```
Out[37]:
```

	county	state	county_state	dates	cases	deaths	us_outbreak
0	Autauga	Alabama	Autauga, Alabama, US	2020-01-22	0	0	0
1	Baldwin	Alabama	Baldwin, Alabama, US	2020-01-22	0	0	0
2	Barbour	Alabama	Barbour, Alabama, US	2020-01-22	0	0	0
3	Bibb	Alabama	Bibb, Alabama, US	2020-01-22	0	0	0
4	Blount	Alabama	Blount, Alabama, US	2020-01-22	0	0	0
...	...	...	...	...	...	...	...
3819901	Teton	Wyoming	Teton, Wyoming, US	2023-03-09	12134	16	1142
3819902	Uinta	Wyoming	Uinta, Wyoming, US	2023-03-09	6406	43	1142
3819903	Unassigned	Wyoming	Unassigned, Wyoming, US	2023-03-09	0	0	1142
3819904	Washakie	Wyoming	Washakie, Wyoming, US	2023-03-09	2755	51	1142
3819905	Weston	Wyoming	Weston, Wyoming, US	2023-03-09	1905	23	1142

3819906 rows × 7 columns

In [39]: `"""
After completing problem 7, run this cell and review the output.
Understand what the as_index parameter does.
"""
df = df_merged.groupby(["us_outbreak", "dates"], as_index=False)["cases"]
df.tail()`

Out[39]:

	us_outbreak	dates	cases
1138	1138	2023-03-05	103646975
1139	1139	2023-03-06	103655539
1140	1140	2023-03-07	103690910
1141	1141	2023-03-08	103755771
1142	1142	2023-03-09	103802702

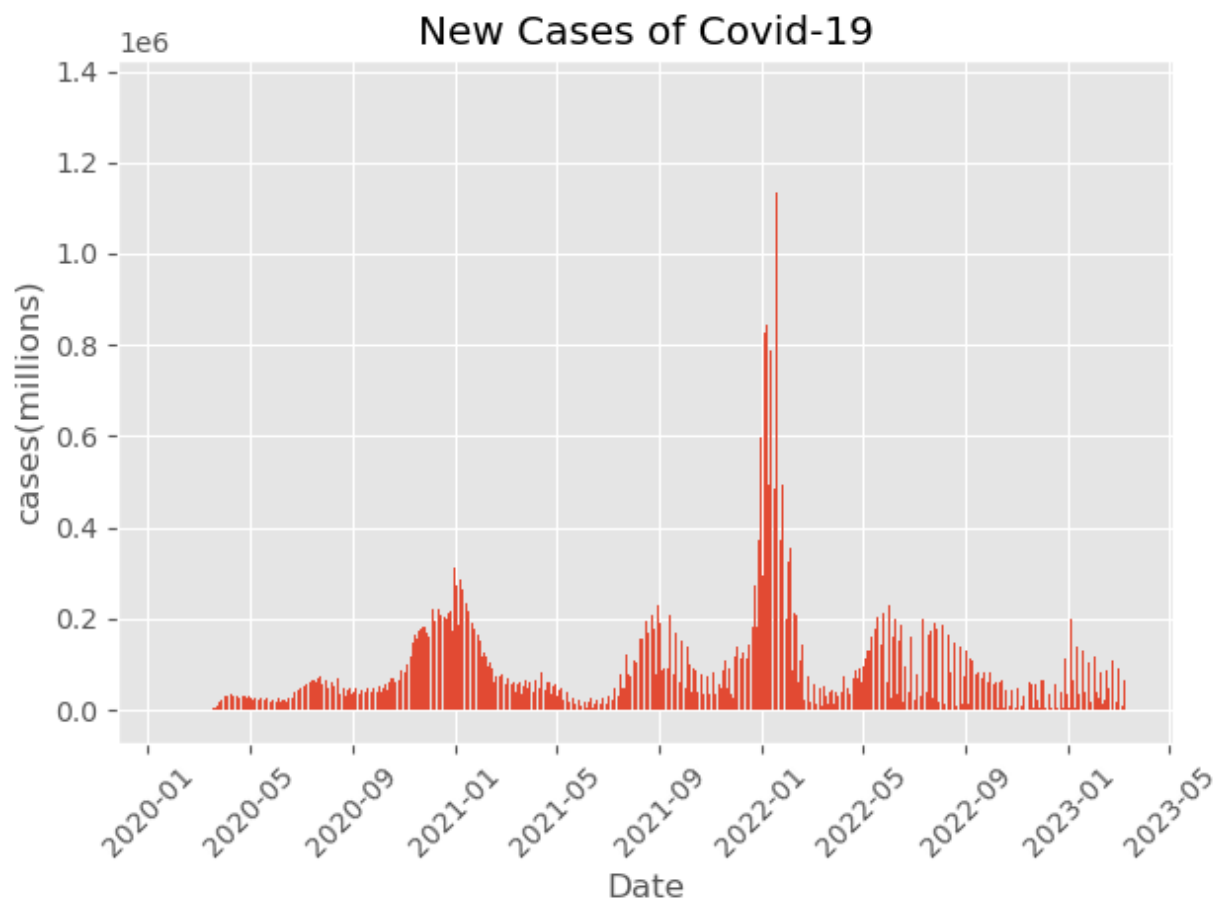
In [40]: `"""Problem 8"""
#Using df from above, create a derived data column using the diff method
#The name of the new column is: new_cases

df["new_cases"] = df['cases'].diff()
df.tail()`

Out[40]:

	us_outbreak	dates	cases	new_cases
1138	1138	2023-03-05	103646975	-3862.0
1139	1139	2023-03-06	103655539	8564.0
1140	1140	2023-03-07	103690910	35371.0
1141	1141	2023-03-08	103755771	64861.0
1142	1142	2023-03-09	103802702	46931.0

```
In [41]: """ Problem 9 """  
#Use plt.bar() function to create a bar plot,  
# have the dates on the x-axis and new_cases on the y-axis.  
#Provide x & y axis labels and an appropriate chart title that  
# explains visual.  
#Rotate the x-axis ticks to 45 using the xticks function.  
# Below are the functions used - complete the code:  
  
plt.figure(1) # include figsize keyword argument  
plt.bar(df["dates"],df["new_cases"])  
plt.xticks(rotation=45) # include rotation keyword argument  
plt.xlabel("Date")  
plt.ylabel("cases(millions)")  
plt.title("New Cases of Covid-19")  
plt.tight_layout()  
plt.show()
```




```
In [44]: """ Problem 10 """  
# Provide the code that shows what the average number of  
# new_cases that were reported in 2021  
new_cases_average_2021 = df[df["dates"].dt.year == 2021]["new_cases"].  
print("average for new cases for 2021 = ", new_cases_average_2021)
```

average for new cases for 2021 = 95035.22465753424

In []: